

MIE377: Financial Optimization Models

Project 2 (Winter 2025)

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1 Introduction

The objective of this project was to develop an automated asset management system that achieves both optimal and consistent investment performance using optimization techniques. The model was trained, validated, and tested using provided data, which included asset prices and factor returns. The final model was evaluated using two key metrics: the Out-of-Sample Sharpe ratio and the average turnover rate. The Sharpe ratio, measured over the entire investment horizon, followed the principle of "higher is better." The average turnover rate was assessed under the guideline that "lower is better." Additionally, the model was subject to a runtime constraint of five minutes.

Five trading algorithms were evaluated based on their performance against the specified criteria. Estimates of expected return (μ vector) and covariance (Q matrix) were refined using factor models, regression, and statistical techniques. The trading algorithm with the highest score was selected as the final model.

The strengths, limitations, and other key insights of the final trading algorithm will be highlighted. Finally, a reflection on what could be done better in the future is provided.

This report is an extension of the Project 1 report with new algorithms and key refinements of model analysis and assessment. The Project 1 report can be viewed in [the Appendix](#).

2 Methodology

This section will introduce the steps in our model selection process. The four new models, and the baseline model, used in testing and their unique considerations will be discussed. The regression models used to derive explanatory factors and estimates of μ and Q will be introduced, as well as their importance. Lastly, this section will provide details regarding the assessment of the algorithms and regression models to provide an overview of the model selection process. Note that we disallowed short-selling in all models.

2.1 Algorithms

2.1.1 Baseline Model: Robust Risk-Return Trade-Off Optimization with Ellipsoidal Uncertainty Set

Our baseline model for comparison, robust risk-return trade-off optimization with an ellipsoidal uncertainty set, was developed for Project 1. It was built on risk-return trade-off optimization [Section 6.2.1] by introducing robustness to the algorithm. Robust risk-return optimization improves traditional optimization methods by addressing the issue of sensitivity to estimation errors in expected returns and the covariance matrix to make portfolios less affected by such inaccuracies. The model is as follows:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \lambda \mathbf{x}^T \mathbf{Q} \mathbf{x} - \left(\boldsymbol{\mu}^T \mathbf{x} - \epsilon \left\| \boldsymbol{\Theta}^{1/2} \mathbf{x} \right\|_2 \right) \\ \text{s.t.} \quad & \mathbf{1}^T \mathbf{x} = 1 \\ & x \geq 0 \end{aligned}$$

A confidence level of 90% was used in the computation of epsilon, which indicates the size of the uncertainty set. A confidence level of 90% results in a larger estimation error range and allows for a balanced portfolio

that manages both risk and return, optimizing the Sharpe ratio. The testing that resulted in a confidence level of 90% is detailed in Project 1 in the Appendix, [Section 6.3.5](#).

2.1.2 Model 1: Risk Parity Optimization

Risk parity optimization maximizes risk diversification, or in other words, equalizes the risk contribution per asset by minimizing the difference in risk contributions. Due to the highly non-linear objective function and non-convexity of this minimization, a retractable form is required. The model can be reformulated as follows:

$$\begin{aligned} \min_y \quad & \frac{1}{2} y^\top Q y - c \sum_{i=1}^N \log(y_i) \\ \text{s.t.} \quad & y_i > 0 \quad \text{for } i = 1, \dots, N \end{aligned}$$

where the optimal weights are extracted with:

$$x = \frac{y}{\sum_{i=1}^N y_i}$$

The objective function of this reformulation is strictly convex because of the second term. Parameter tuning was unnecessary for this formulation as the optimal weights are independent of c .

2.1.3 Model 2: CVaR Optimization

Conditional value-at-risk measures the expected value of losses greater than or equal to VaR. This model minimizes the downside risk only, as opposed to minimizing all deviations in value even when they improve performance. Similarly to risk parity, a retractable form is required because value-at-risk and probability density functions are challenging to solve analytically. The convex CVaR optimization model appears as follows:

$$\begin{aligned} \min_{x, z, \gamma} \quad & \gamma + \frac{1}{(1-\alpha)S} \sum_{s=1}^S z_s \\ \text{s.t.} \quad & z_s \geq 0, \quad \text{for } s = 1, \dots, S \\ & z_s \geq -r_s^\top x - \gamma, \quad \text{for } s = 1, \dots, S \\ & \mathbf{1}^\top x = 1 \\ & \mu^\top x \geq R \\ & x \geq 0 \end{aligned}$$

A confidence level of 95% was used in the optimization. A confidence level of 95% in CVaR measures the average loss that occurs in the worst 5% of scenarios. The testing that resulted in a confidence level of 95% is detailed in [Section 3.4.1](#).

2.1.4 Model 3: Robust CVaR Optimization

Robust conditional value-at-risk is a robustified version of Model 2. The advantages of robustness in CVaR are similar to those described in the baseline model above. The robust version of CVaR optimization is as follows:

$$\begin{aligned}
\min_{x, z, \gamma} \quad & \gamma + \frac{1}{(1 - \alpha)S} \sum_{s=1}^S z_s \\
\text{s.t.} \quad & z_s \geq 0, \quad \text{for } s = 1, \dots, S \\
& z_s \geq -r_s^\top x - \gamma, \quad \text{for } s = 1, \dots, S \\
& \mathbf{1}^\top x = 1 \\
& \mu_{\text{robust}}^\top x \geq R \\
& x \geq 0
\end{aligned}$$

For consistency purposes, a confidence level of 95% was also used in Model 3.

2.1.5 Model 4: Sharpe Ratio Optimization

Sharpe ratio optimization, as the name suggests, maximizes the ex-ante Sharpe ratio. Unsurprisingly, the objective function is highly non-linear, resulting in the need for a convex reformulation, which appears as follows:

$$\begin{aligned}
\min_{\mathbf{y}, \kappa} \quad & \mathbf{y}^\top Q \mathbf{y} \\
\text{s.t.} \quad & (\boldsymbol{\mu} - \bar{r}_f)^\top \mathbf{y} = 1 \\
& \mathbf{1}^\top \mathbf{y} = \kappa \\
& A \mathbf{y} \leq b \cdot \kappa \\
& \kappa \geq 0 \\
& \mathbf{y} \geq 0
\end{aligned}$$

where the optimal weights are extracted with:

$$x_j^* = \frac{y_j^*}{\kappa^*} = \frac{y_j^*}{\sum_{i=1}^n y_i^*}$$

No parameter tuning was necessary for Sharpe ratio optimization.

2.2 Model Considerations

2.2.1 Model choices

Our motivation for utilizing the above models stem from the concepts learned in this course. We wanted to demonstrate our ability to successfully apply the algorithms using our practice from the lab sessions. Additionally, we wanted to make accurate and insightful conclusions to expand our knowledge of these models in a practical setting. Finally, we were eager to use these new models to achieve better results than in Project 1.

2.2.2 Short Selling

The goal is to maximize the portfolio Sharpe ratio while minimizing the average turnover rate, essentially crafting a theoretically optimal portfolio. Disallowing short selling can decrease the average turnover rate and reduce downside risk which boosts the Sharpe ratio. Furthermore, disallowing short selling can enhance out-of-sample performance, a key component of this project. Hence, all of the algorithms investigated disallow short selling.

Note that some models force long-only portfolios. Risk-parity optimization is a key example of this. The natural logarithm in the retractable formulation disallows short selling by construction.

2.2.3 Data Cleaning and Validation

We chose to use all five years of data rather than subsetting it to avoid excessive overfitting. Reducing the dataset to less than five years would result in a model specific to the small dataset, affecting its ability to perform on new, unseen data.

2.3 Regularization Techniques and Factor Models

Regularization techniques were used to determine the factors that explain the most variation in the model. These best factors were used in forming a factor model that estimates μ and Q , which were used in calculating the Sharpe ratio and average turnover rate in the models. Additionally, ubiquitous factor models can be used for the same purpose. It is crucial to use factor models to estimate μ and Q because they are typically more accurate and less noisy than sample mean and covariance. Factor models also allow for better performance on new datasets as the estimates are more generalizable. In this section, the regularization techniques used to obtain these estimates, and the reasoning behind their forms, will be introduced, in addition to well-known factor models. Furthermore, principal component analysis will be discussed.

2.3.1 Best Subset Selection (BSS)

The penalized form of BSS uses the l_0 -norm, which is structured as follows:

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \|\mathbf{B}_i\|_0$$

$$\|\mathbf{B}_i\|_0 = \mathcal{K}_{\alpha_i \neq 0} + \sum_{k=1}^p \mathcal{K}_{\beta_k \neq 0}$$

BSS selects the most important factors by setting a penalty on the number of non-zero coefficients which makes the norm values smaller. However, the penalized form does not allow explicit control on the number of non-zero elements. BSS promotes true sparsity, allowing several coefficients to be zero at optimality. However, the l_0 -norm is not convex, so BSS is not convex, resulting in BSS performing best on mixed-integer programming models with binary variables. Nevertheless, for completeness, insight, and curiosity, BSS was tested with the five algorithms listed previously.

2.3.2 Least Absolute Shrinkage and Selection Operator (LASSO)

The penalized form of LASSO uses the l_1 -norm and appears as follows:

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \|\mathbf{B}_i\|_1$$

$$\|\mathbf{B}_i\|_1 = |\alpha_i| + \sum_{k=1}^p |\beta_{ik}|$$

LASSO penalizes the sum of absolute values of the factors which also makes the norm values smaller. Similar to BSS, LASSO promotes true sparsity resulting in zero-valued coefficients. The l_2 -norm is a convex function, so LASSO is also convex, which allows for efficiency when solving.

2.3.3 Ridge Regression (RR)

RR, the final regularization technique, utilizes the l_2 -norm. The penalized form is:

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \|\mathbf{B}_i\|_2^2$$
$$\|\mathbf{B}_i\|_2 = \left((\alpha_i)^2 + \sum_{k=1}^p (\beta_{ik})^2 \right)^{1/2}$$

Uniquely, RR penalizes the Euclidean norm. However, RR does not promote true sparsity and instead attempts to shrink the coefficients to zero. However, the l_2 -norm is continuous, convex, and smooth, and allows a closed-form solution.

2.3.4 Fama-French Three Factor Model

This Three Factor Model uses the market factor, size factor, and value factor, and is given by the following:

$$r_i - r_f = \alpha_i + \beta_{im}(f_m - r_f) + \beta_{is}f_s + \beta_{iv}f_v + \varepsilon_i$$

Note that the factors are correlated in the Three-Factor Model as they are subsets of market data.

2.3.5 Capital Asset Pricing Model (CAPM)

The final factor model is CAPM which is a one-factor model of the market excess return. CAPM is known by the following:

$$r_i - r_f = \beta_{im}(f_m - r_f) + \varepsilon_i$$

2.3.6 Ordinary Least Squares Regression (OLS)

A simple OLS regression with the eight provided factors was also used. OLS regression minimizes the sum of squared residuals, similar to BSS, LASSO, and RR, but is subject to no constraints. OLS regression is given by:

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2$$

2.3.7 Principal Component Analysis (PCA)

PCA is a linear transformation tool that extracts a set of factors from asset returns. A subset of these factors is selected as the principal components (PCs) to reduce dimensionality, where each PC explains a certain amount of variability in the model. The following model was used:

$$P = \bar{R}\Gamma$$

where $\bar{R}$ is the centered data matrix of observed asset returns, defined as $\bar{R} = R - \mathbf{1}\bar{r}^\top$, and Γ is from the eigen-decomposition of $Q = \Gamma\Lambda\Gamma^\top$. PCA is largely beneficial for noise reduction and well-suited for avoiding overfitting. It reduces noise in the estimation of the covariance matrix by focusing on capturing only the top few principal components that explain the most variance.

2.4 Factor model considerations

The regularization techniques and factor models described above were chosen to estimate μ and Q for reasons similar to those in *2.2.1 Model Choices*. We wanted to apply a range of techniques learned in this course

to develop a practical understanding of how each technique worked. This project served as an excellent opportunity for us to see the results of these methods, which expanded our understanding and built our intuition.

The regularization techniques can be used in their penalized or constrained forms. The LASSO and RR models are equivalent in their penalized or constrained form, but BSS is not. The penalized form of BSS was used because it is more flexible than its constrained form. The constrained form requires a manually selected cardinality constraint, which limits computational feasibility. Rather than arbitrarily estimating a cardinality constraint, the penalized form uses automatic factor selection which is better for large datasets where strict control is not necessary. It is better to adjust the penalty to make the norm smaller to obtain a balance between the number of factors and accuracy.

Finally, a subset of the factors extracted with PCA must be selected. A thorough analysis was conducted to determine the optimal number of PCs to select and will be discussed later on.

2.5 Assessment of Models

In this project, we expected better model performance compared to Project 1 based on our intuition and newly developed skills. Hence, to properly evaluate the models, we used our best-performing model from Project 1 as our baseline for comparison. All of the algorithms, including the baseline model, were run on each dataset with the regression techniques and factor models. The optimal number of PCs was determined by running simulations for every possible number of PCs on each dataset and selecting the number of PCs that maximized the Sharpe ratio for each unique algorithm. The Sharpe ratios, average turnover rates, and runtimes were recorded. The ideal combinations of algorithms and techniques to estimate μ and Q were determined by averaging the runtime, Sharpe ratio, and turnover rate across the three datasets. The technique with the highest average Sharpe ratio was selected as the best combination with that algorithm. Finally, each combination was allocated a score using the following formula:

$$0.8 \times \text{Sharpe ratio} + 0.2 \times (1 - \text{Average turnover rate}) \quad (1)$$

The formula was created to score each model according to the weighting of the assessment criteria: 80% of the performance being attributed to the Sharpe ratio and 20% being attributed to the turnover rate.

The model combination with the highest score was deemed the final, best-performing model.

3 Analysis

3.1 Robust Risk-Return Trade-Off with Ellipsoidal Uncertainty Set

First, the performance of the baseline model was assessed using all three datasets and all regularization techniques. The results of the tests are presented in Tables 3.1.1, 3.1.2, and 3.1.3 below.

Table 3.1.1. Baseline Model: Dataset 1

	OLS	Three-Factor	CAPM	RR	LASSO	BSS	PCA (1 PC)
Time (s)	0.6082	0.6166	0.6463	0.6142	0.9900	0.6657	13.8520
Sharpe Ratio	0.1708	0.1779	0.1810	0.1760	0.1823	0.1781	0.1796
Avg. Turnover	0.5258	0.4888	0.4910	0.5105	0.4881	0.4929	0.5172

Table 3.1.2. Baseline Model: Dataset 2

	OLS	Three-Factor	CAPM	RR	LASSO	BSS	PCA (1 PC)
Time (s)	0.7531	0.7384	0.7261	0.7471	0.9603	0.8587	14.9489
Sharpe Ratio	0.1014	0.0963	0.1124	0.1072	0.1050	0.1061	0.0948
Avg. Turnover	0.6517	0.6580	0.6569	0.6466	0.6529	0.6460	0.6706

Table 3.1.3. Baseline Model: Dataset 3

	OLS	Three-Factor	CAPM	RR	LASSO	BSS	PCA (1 PC)
Time (s)	0.7616	0.7362	0.7479	0.7794	0.9749	0.8332	13.5633
Sharpe Ratio	0.1672	0.1609	0.1564	0.1630	0.1563	0.1507	0.1802
Avg. Turnover	0.6348	0.5722	0.5919	0.6196	0.6118	0.6202	0.6115

Computation times were similar across all datasets and regularization techniques, except for PCA which maintained a consistently higher computation time with all datasets. Running the model with dataset 1 consistently resulted in the highest Sharpe ratios, with dataset 3 closely following. Dataset 1 also consistently had the lowest average turnover rates, with dataset 3 closely following. Dataset 2 consistently had the lowest Sharpe ratios and highest turnover rates. The differences in performance across datasets may be attributed to variations in the underlying asset characteristics, return distributions, or correlations within each dataset, which can impact both risk-adjusted returns and trading activity.

After obtaining the model results for all three datasets, the average values across the datasets were computed. The mean values are presented in Table 3.1.4 below.

Table 3.1.4. Baseline Model: Average

	OLS	Three-Factor	CAPM	RR	LASSO	BSS	PCA
Time (s)	0.7076	0.6971	0.7068	0.7136	0.9751	0.7859	14.1214
Sharpe Ratio	0.1464	0.1450	0.1499	0.1487	0.1479	0.1450	0.1515
Avg. Turnover	0.6041	0.5730	0.5799	0.5922	0.5843	0.5864	0.5998

The best performing regularization technique was PCA (highlighted in Table 3.1.4), with the highest Sharpe ratio of 0.1515.

The score for the robust optimization model using PCA was then calculated using the formula discussed in [Section 2.5](#):

$$0.8 \times 0.1515 + 0.2 \times (1 - 0.5998) = \mathbf{0.2013}$$

3.1.1 Robust Optimization - PC Sensitivity Analysis

A sensitivity analysis was conducted on the PCA regularization technique with the robust optimization model to determine the optimal number of principal components. The optimal number of principal components was determined by aiming to maximize the Sharpe ratio. The results of the sensitivity analysis can be seen in Figures 3.1.1.1, 3.1.1.2, and 3.1.1.3 below.

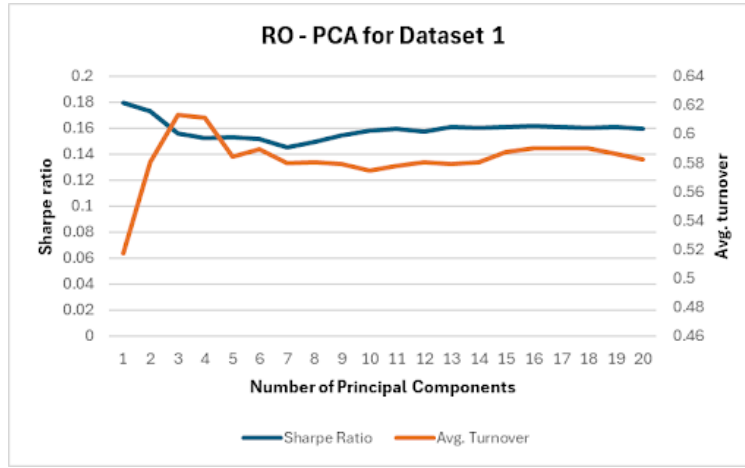


Figure 3.1.1.1. Results of PC sensitivity analysis for baseline model using dataset 1.

Using a single principal component achieves the highest Sharpe ratio for dataset 1.

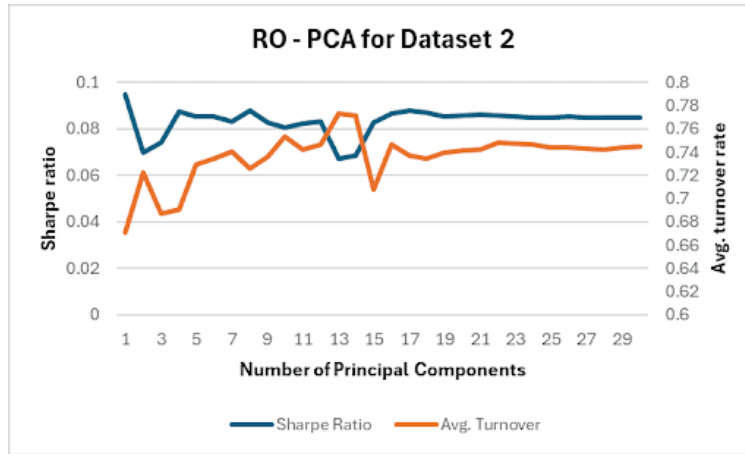


Figure 3.1.1.2. Results of PC sensitivity analysis for baseline model using dataset 2.

Using a single principal component achieves the highest Sharpe ratio for dataset 2.

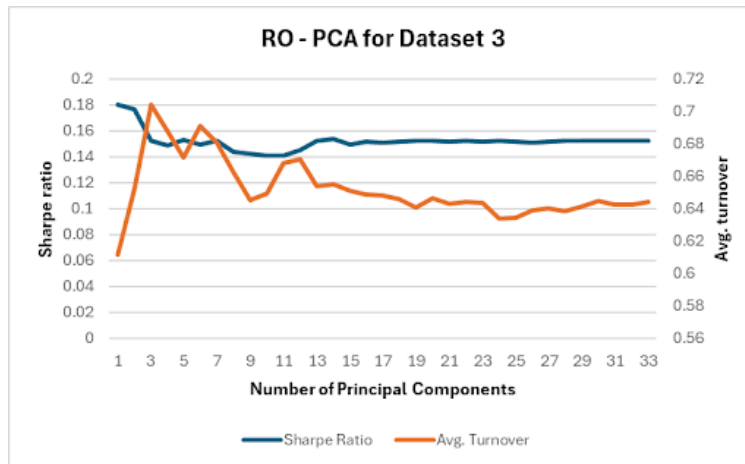


Figure 3.1.1.3. Results of PC sensitivity analysis for baseline model using dataset 3.

Using a single principal component achieves the highest Sharpe ratio for dataset 3.

Notably, the robust optimization model using PCA performed optimally on all three datasets using only one PC. It is plausible that using a small subset of PCs performs well, as selecting too many PCs may result in overfitting.

3.2 Risk Parity Optimization

Next, the performance of the risk parity optimization model was assessed using all three datasets and all regularization techniques. The results of the tests are presented in Tables 3.2.1, 3.2.2, and 3.2.3 below.

Table 3.2.1. Risk Parity: Dataset 1

	OLS	Three-Factor	CAPM	RR	LASSO	BSS	PCA (7 PCs)
Time (s)	2.6773	2.6941	2.6615	2.6877	3.8716	2.9349	11.8060
Sharpe Ratio	0.1824	0.1823	0.1796	0.1823	0.1795	0.1800	0.1864
Avg. Turnover	0.1340	0.1317	0.1291	0.1344	0.1372	0.1337	0.1438

Table 3.2.2. Risk Parity: Dataset 2

	OLS	Three-Factor	CAPM	RR	LASSO	BSS	PCA (1 PC)
Time (s)	2.5765	2.4377	2.4373	2.5523	2.8986	2.5855	10.8474
Sharpe Ratio	0.1254	0.1252	0.1339	0.1253	0.1256	0.1255	0.1307
Avg. Turnover	0.1021	0.1033	0.1075	0.1032	0.1031	0.1033	0.1087

Table 3.2.3. Risk Parity: Dataset 3

	OLS	Three-Factor	CAPM	RR	LASSO	BSS	PCA (9 PCs)
Time (s)	2.5455	2.4618	2.4254	2.4777	2.8926	2.5267	11.6917
Sharpe Ratio	0.1813	0.1791	0.1787	0.1801	0.1808	0.1798	0.1916
Avg. Turnover	0.1503	0.1548	0.1539	0.1549	0.1494	0.1486	0.1540

Computation times across all three datasets were very similar for the risk parity optimization model, except for PCA which again maintained a consistently higher computation time on all the datasets. Dataset 1 and dataset 3 consistently resulted in similar Sharpe ratios, with dataset 2 consistently resulting in the lowest Sharpe ratios. Dataset 2 consistently had the lowest average turnover rates, dataset 1 had the second lowest average turnover rates, and dataset 3 had the highest average turnover rates. The differences can be attributed to the same reasons as the rest of the models.

After obtaining the model results for all three datasets, the average values across the datasets were computed. The mean values are presented in Table 3.2.4 below.

Table 3.2.4. Risk Parity: Average

	OLS	Three-Factor	CAPM	RR	LASSO	BSS	PCA
Time (s)	2.5998	2.5312	2.5081	2.5726	3.2209	2.6824	11.5235
Sharpe Ratio	0.1630	0.1622	0.1641	0.1625	0.1620	0.1617	0.1695
Avg. Turnover	0.1288	0.1299	0.1301	0.1308	0.1299	0.1286	0.1355

Across the three datasets, the best-performing regularization technique on average was PCA (highlighted in Table 3.2.4), with the highest Sharpe ratio of 0.1695.

The score for the risk parity optimization model using PCA was then calculated using the formula discussed in [Section 2.5](#):

$$0.8 \times 0.1695 + 0.2 \times (1 - 0.1355) = \mathbf{0.308536}$$

3.2.1 Risk Parity Optimization - PC Sensitivity Analysis

A sensitivity analysis was completed on the PCA regularization technique with the risk parity optimization model to determine the optimal number of principal components. The results of the sensitivity analysis can be seen in Figures 3.2.1.1, 3.2.1.2, and 3.2.1.3 below.

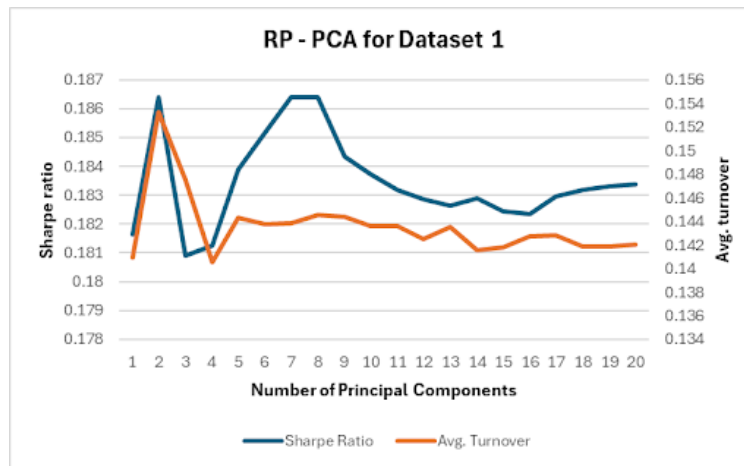


Figure 3.2.1.1. Results of PC sensitivity analysis for risk parity model using dataset 1.

Using 7 principal components achieves the highest Sharpe ratio for dataset 1.

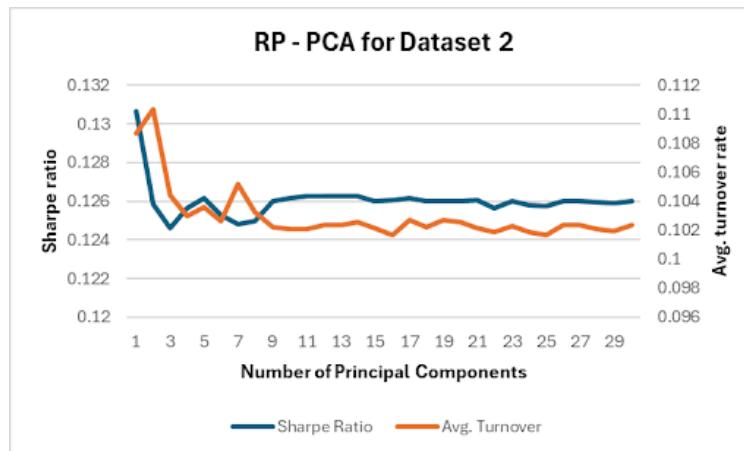


Figure 3.2.1.2. Results of PC sensitivity analysis for risk parity model using dataset 2.

Using a single principal component achieves the highest Sharpe ratio for dataset 2.

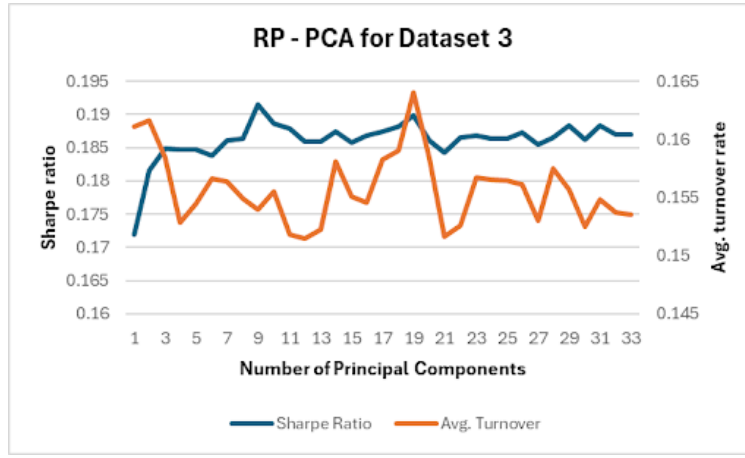


Figure 3.2.1.3. Results of PC sensitivity analysis for risk parity model using dataset 3.

Using 9 principal components achieves the highest Sharpe ratio for dataset 3.

Comparably to the baseline model, it is reasonable that using a small subset of PCs performs well, as selecting too many PCs may result in overfitting.

3.3 Sharpe Ratio Optimization

Furthermore, the performance of the Sharpe ratio optimization model was assessed using all three datasets and all regularization techniques. The results of the tests are presented in Tables 3.3.1, 3.3.2, and 3.3.3 below.

Table 3.3.1. Sharpe Ratio: Dataset 1

	OLS	Three-Factor	CAPM	RR	LASSO	BSS	PCA (11 PCs)
Time (s)	0.7030	0.6827	0.6739	0.6856	0.9550	0.7721	16.5085
Sharpe Ratio	0.2094	0.2014	0.2052	0.2184	0.2198	0.2178	0.2082
Avg. Turnover	0.5272	0.4863	0.4575	0.5229	0.5005	0.5225	0.5659

Table 3.3.2. Sharpe Ratio: Dataset 2

	OLS	Three-Factor	CAPM	RR	LASSO	BSS	PCA (4 PCs)
Time (s)	1.2880	0.6997	1.2708	1.3100	0.6903	1.4185	12.2626
Sharpe Ratio	0.1034	0.0978	0.0978	0.1084	0.1024	0.1074	0.1045
Avg. Turnover	0.6997	0.6966	0.6822	0.6903	0.7180	0.7050	0.6901

Table 3.3.3. Sharpe Ratio: Dataset 3

	OLS	Three-Factor	CAPM	RR	LASSO	BSS	PCA (28 PCs)
Time (s)	2.5215	1.5111	1.4693	1.3777	1.8312	1.5133	11.4790
Sharpe Ratio	0.1867	0.1857	0.1813	0.1728	0.1779	0.1646	0.1968
Avg. Turnover	0.5570	0.5121	0.5035	0.5623	0.5457	0.5648	0.6068

Computation times across all three datasets consistently differed slightly; with dataset 1 generally taking the shortest amount of time, dataset 2 taking the second place, and dataset 3 taking the longest amount of time.

However, they did not differ significantly. Again, running the model with PCA maintained a consistently higher computation time with all the datasets. Dataset 1 and dataset 3 had similarly high Sharpe ratios, with dataset 2 having the lowest; similar to risk parity optimization. Dataset 1 and dataset 3 also had similar turnover rates, and dataset 2 had consistently higher turnover rates than its counterparts. The differences can be attributed to the same reasons as the rest of the models.

After obtaining the model results for all three datasets, the average values across the datasets were computed. The mean values are presented in Table 3.3.4 below.

Table 3.3.4. Sharpe Ratio: Average

	OLS	Three-Factor	CAPM	RR	LASSO	BSS	PCA
Time (s)	1.5042	0.9645	1.1380	1.1244	1.1588	1.2346	13.4167
Sharpe Ratio	0.1665	0.1616	0.1614	0.1665	0.1667	0.1632	0.1698
Avg. Turnover	0.5946	0.5650	0.5478	0.5919	0.5881	0.5974	0.6210

Across the three datasets, the best-performing regularization technique on average was PCA (highlighted in Table 3.3.4), with the highest Sharpe ratio of 0.1698.

The score for the Sharpe ratio based optimization model using PCA was then calculated using the formula discussed in Section 2.5:

$$0.8 \times 0.1698 + 0.2 \times (1 - 0.6210) = \mathbf{0.2117}$$

3.3.1 Sharpe Ratio Optimization - PC Sensitivity Analysis

A sensitivity analysis was completed on the PCA regularization technique with the Sharpe ratio based optimization model to determine the optimal number of principal components. The results of the sensitivity analysis can be seen in Figures 3.3.1.1 - 3.3.1.3 below.

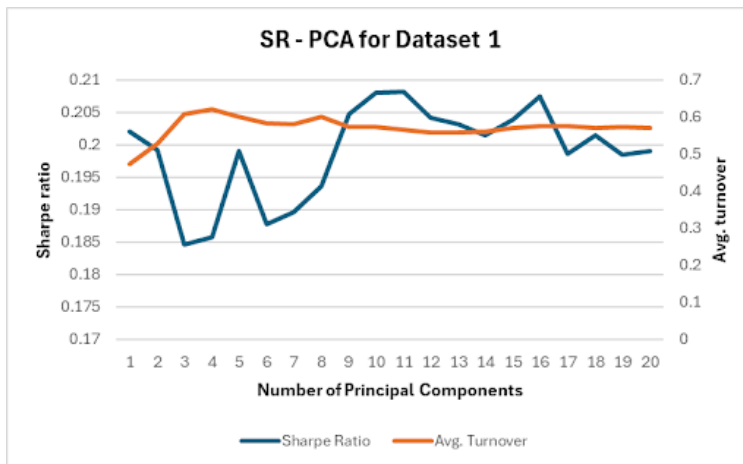


Figure 3.3.1.1. Results of PC sensitivity analysis for Sharpe ratio model using dataset 1.

Using 11 principal components achieves the highest Sharpe ratio for dataset 1.

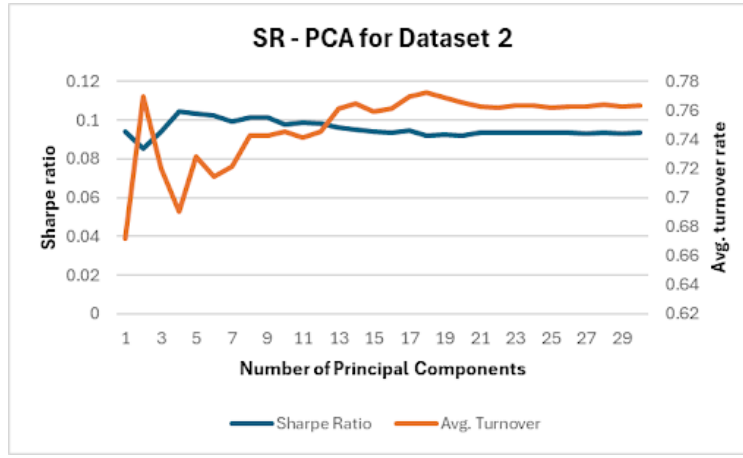


Figure 3.3.1.2. Results of PC sensitivity analysis for Sharpe ratio model using dataset 2.

Using 4 principal components achieves the highest Sharpe ratio for dataset 2.

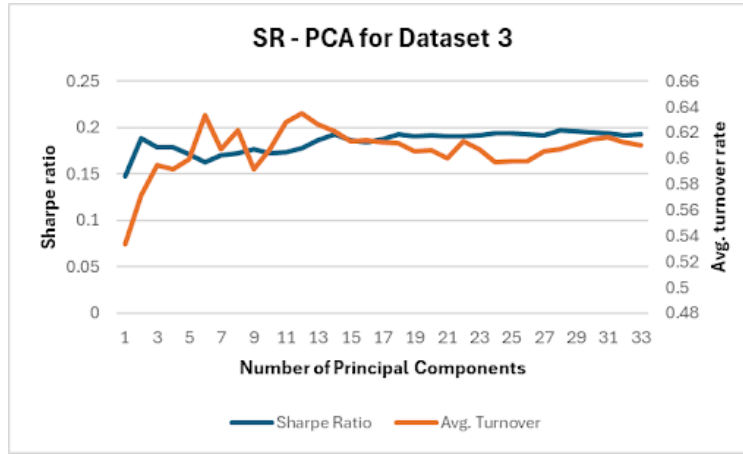


Figure 3.3.1.3. Results of PC sensitivity analysis for Sharpe ratio model using dataset 3.

Using 28 principal components achieves the highest Sharpe ratio for dataset 3.

The Sharpe ratio optimization model applied to Dataset 3 yields the highest optimal number of principal components across all models and datasets, with a value of 28. This introduces risk of overfitting.

3.4 CVaR Optimization

The performance of the CVaR optimization was assessed using all three datasets. The model assessment was based solely on historical returns and factor returns. Given the availability of three datasets, simulating returns using factor models is unlikely to provide significant additional benefit. The results of the tests are presented in Tables 3.4.1 - 3.4.3 below.

Table 3.4.1. CVaR: Dataset 1

Historical Return/FactRet	
Time (s)	1.0181
Sharpe Ratio	0.2205
Avg. Turnover	0.6509

Table 3.4.2. CVaR: Dataset 2

	Historical Return/FactRet
Time (s)	1.1435
Sharpe Ratio	0.0959
Avg. Turnover	0.6372

Table 3.4.3. CVaR: Dataset 3

	Historical Return/FactRet
Time (s)	1.1039
Sharpe Ratio	0.2239
Avg. Turnover	0.5436

Computation times were very similar across all three datasets. Dataset 1 and dataset 3 resulted in similarly high Sharpe ratios, with dataset 3's Sharpe ratio being slightly higher. Dataset 2 had the lowest Sharpe ratio. Dataset 3 had the lowest average turnover rate, dataset 2 had the second lowest average turnover rate, and dataset 1 had a slightly higher turnover rate compared to dataset 2.

After obtaining the model results for all three datasets, the average values across the datasets were computed. The mean values are presented in Table 3.4.4 below.

Table 3.4.4. CVaR: Average

	Historical Return/FactRet
Time (s)	1.0885
Sharpe Ratio	0.1801
Avg. Turnover	0.6106

The score for the CVaR optimization model was then calculated using the average values across all three datasets and the formula discussed in [Section 2.5](#):

$$0.8 \times 0.1801 + 0.2 \times (1 - 0.6106) = \mathbf{0.2220}$$

3.4.1 CvaR Parameter Tuning - α

Parameter tuning was also done on the confidence level, α . Different values of α were tested on the CVaR optimization model using dataset 1. The results of the tests can be seen in Table 3.4.1.1 below.

Table 3.4.1.1. Parameter Estimation of α (Using Regular CVaR, Dataset 1)

α	0.90	0.91	0.92	0.93	0.94	0.95	0.96	0.97	0.98	0.99
Sharpe Ratio	0.1462	0.1539	0.1568	0.1646	0.1646	0.1646	0.1646	0.1646	0.1646	0.1646
Turnover	3.0861	3.1084	3.1697	3.1724	3.1724	3.1724	3.1724	3.1724	3.1724	3.1724

A lower α resulted in a lower Sharpe ratio, but increasing α past 0.95 did not increase the Sharpe ratio any further. As such, 0.95 was chosen as the CVaR optimization model's risk aversion parameter. A confidence level that is too high may result in lower returns and worse performance in unseen datasets, as it results in an extremely risk-averse model. $\alpha = 0.95$ was determined to be a good balance of allowing risk and prioritizing returns because it resulted in the same Sharpe ratio and turnover rate as $\alpha = 0.96, 0.97, 0.98, 0.99$.

3.5 Robust CVaR Optimization

Finally, the performance of the robust CVaR optimization was assessed using all three datasets. As with the original CVaR optimization model, the model assessment was based solely on historical returns and factor returns. The results of the tests are presented in Tables 3.5.1 - 3.5.3 below.

Table 3.5.1. Robust CVaR: Dataset 1

	Historical Return/FactRet
Time (s)	1.0963
Sharpe Ratio	0.2043
Avg. Turnover	0.6639

Table 3.5.2. Robust CVaR: Dataset 2

	Historical Return/FactRet
Time (s)	1.1724
Sharpe Ratio	0.1142
Avg. Turnover	0.7454

Table 3.5.3. Robust CVaR: Dataset 3

	Historical Return/FactRet
Time (s)	1.1632
Sharpe Ratio	0.2115
Avg. Turnover	0.5652

Computation times were very similar across all three datasets. As with the original CVaR optimization model, dataset 1 and dataset 3 resulted in similarly high Sharpe ratios, with dataset 3's Sharpe ratio being slightly higher. Dataset 2 had the lowest Sharpe ratio. Dataset 3 had the lowest average turnover rate, dataset 1 had the second lowest average turnover rate, and dataset 2 had the highest turnover rate.

After obtaining the model results for all three datasets, the average values across the datasets were computed. The mean values are presented in Table 3.5.4 below.

Table 3.5.4. Robust CVaR: Average

	Historical Return/FactRet
Time (s)	1.1440
Sharpe Ratio	0.1766
Avg. Turnover	0.6582

The score for the robust CVaR optimization model was then calculated using the average values across all three datasets and the formula discussed in [Section 2.5](#):

$$0.8 \times 0.1766 + 0.2 \times (1 - 0.6582) = \mathbf{0.2097}$$

Note that $\alpha = 0.95$ was used for the robust CVaR optimization model for the same reasons detailed in [Section 3.4.1](#). Additionally, it was also found that changing the robustness factor did not change the performance of the model, so it was kept at a standard 0.5.

3.6 Final Model Selection

The scores for each model combination are displayed in Table 3.6.1 below.

Table 3.6.1. Best Models Based on Sharpe Ratio

	Robust – PCA	Risk Parity – PCA	Sharpe – PCA	CVaR	Robust CVaR
Score	0.201252	0.308536	0.211684	0.221986	0.209673

Risk parity optimization with PCA achieved the highest score making it the final selected model. The model's wealth evolution and asset weighting graphs for all three datasets are displayed in Figures 3.6.1-3.6.6 below.

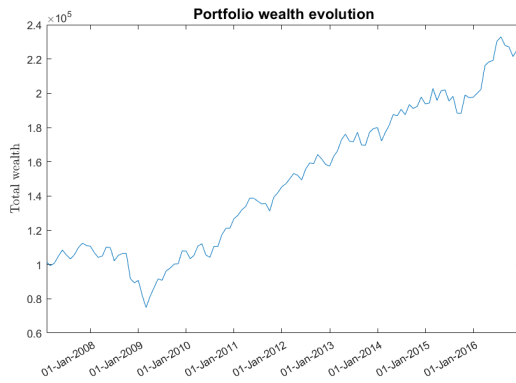


Figure 3.6.1. Wealth evolution for dataset 1.

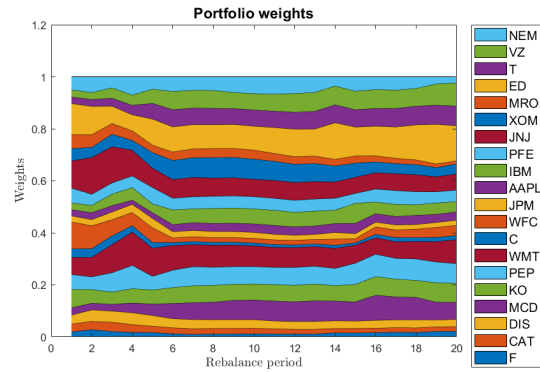


Figure 3.6.2. Asset weightings for dataset 1.

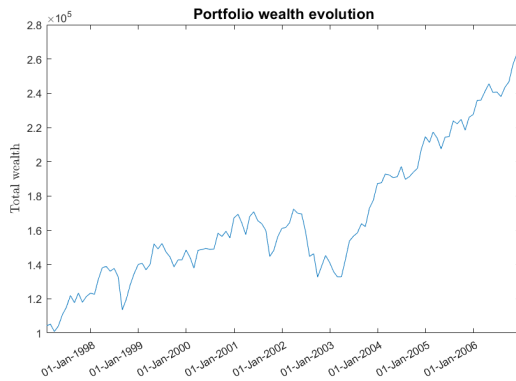


Figure 3.6.3. Wealth evolution for dataset 2.

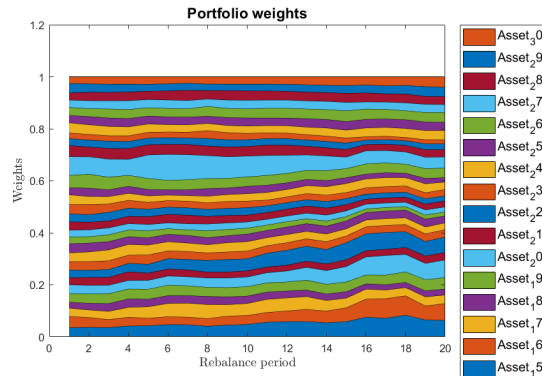


Figure 3.6.4. Asset weightings for dataset 2.

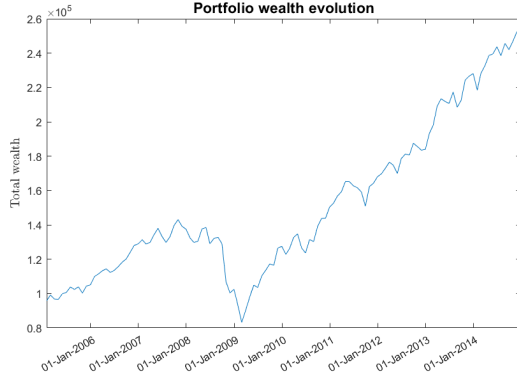


Figure 3.6.5. Wealth evolution for dataset 3.

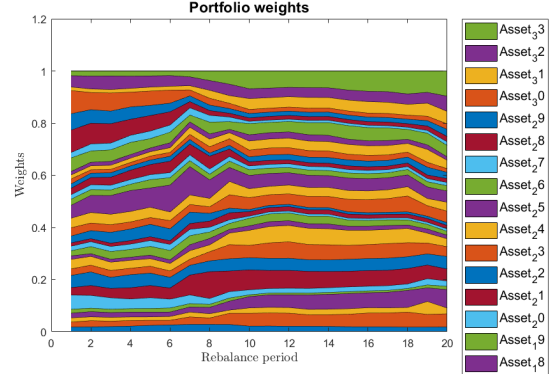


Figure 3.6.6. Asset weightings for dataset 3.

4 Discussion

4.1 Final Model Strengths

Risk-parity optimization with PCA estimation was chosen as the best strategy for automated asset management because it produced the lowest turnover rates while maintaining relatively high Sharpe ratios across datasets. Although Sharpe ratio optimization resulted in the highest Sharpe ratios, its turnover rates were extremely high, making it a suboptimal asset management strategy. Additionally, risk parity naturally leads to lower turnover because it allocates risk rather than capital, stabilizing the portfolio weights even when expected returns fluctuate. This technique diversifies risk and ensures that no specific asset dominates the portfolio risk. As a result, it is not heavily sensitive to errors in estimating sample covariance and is not dependent on sample mean.

Furthermore, PCA estimation helps prevent overfitting of Q when historical data is limited or unstable. PCA only extracts dominant patterns (principal components) and filters out noise in datasets through dimensionality reduction. This results in more stable and robust estimates of Q and μ . Consequently, PCA estimates improve out-of-sample performance compared to traditional regression-based techniques analyzed above. Overall, risk parity with PCA is an effective asset management strategy as it typically performs consistently during both bull and bear markets.

4.2 Limitations of Final Model

As mentioned previously, risk parity ignores μ and may therefore under-allocate to assets that consistently have higher historical returns in the portfolio. As a result, this model does not necessarily optimize expected return and can miss favorable risk-return opportunities. Furthermore, PCA may oversimplify the market structure through dimensionality reduction. The assumption that a few principal components capture all meaningful variance may not hold during periods of highly volatile market conditions. Like the other models we explored in this study, risk parity and PCA estimation rely on historical data. Consequently, it assumes that volatility and correlation between assets remain stable over time. This makes the strategy less responsive to sudden market shifts, making it less robust in dynamic environments.

5 Conclusion

This report evaluated multiple portfolio optimization strategies across three datasets using various regularization techniques, with performance measured by Sharpe ratio and average turnover rate. Among the approaches tested, risk parity optimization with PCA emerged as the most effective strategy based on a

scoring metric that balanced risk-adjusted return and portfolio turnover.

Risk parity with PCA consistently delivered high Sharpe ratios with low turnover, demonstrating both robustness and efficiency. PCA’s dimensionality reduction played a crucial role in improving out-of-sample performance by filtering out noise and preventing overfitting, especially when historical data was limited. This made the combined model resilient to estimation errors in both expected returns and covariances.

While the strategy’s reliance on historical data and omission of mean return pose limitations—particularly in dynamic market environments or when return forecasts are informative—its overall stability and consistency make it a strong candidate for automated asset management. Ultimately, risk parity with PCA strikes a practical balance between diversification, risk control, and computational efficiency, making it a valuable tool for long-term portfolio allocation under uncertainty.

In the future, model performance could be further enhanced by incorporating dynamic market indicators. Investigating the responses of the models under bull and bear markets, and using external datasets outside of those provided, could allow greater insight into the robustness and efficiency of the models.

6 Appendix (Project 1)

6.1 Introduction

The objective of this project was to develop an automated asset management system that achieves both optimal and consistent investment performance using mean-variance optimization (MVO). The model was trained, validated, and tested using provided data, which included asset prices and factor returns.

The final model was evaluated using two key metrics: the Out-of-Sample Sharpe ratio and the average turnover rate. The Sharpe ratio, measured over the entire investment horizon, followed the principle of “higher is better.” The average turnover rate was assessed under the guideline that “lower is better.” Additionally, the model was subject to a runtime constraint of five minutes.

Four trading algorithms were evaluated based on their performance against the specified criteria. Estimates of expected return (μ vector) and covariance (Q matrix) were refined using factor models. In addition to factor model estimates, μ and Q were directly estimated from asset returns and tested within the models. The final trading algorithm was selected based on the specified criteria.

Strengths, limitations, and other key insights of the final trading algorithm will be highlighted. Finally, a reflection on what could be done better in the future is detailed.

6.2 Methodology

This section will introduce the steps in our model selection process. The four MVO models used in testing, and their unique considerations, will be discussed. The regression models used to derive explanatory factors and estimates of μ and Q will be introduced, as well as their importance. Lastly, this section will provide details regarding the assessment of the MVO algorithms and regression models to provide an overview of the model selection process. Note that we allowed short-selling in all models.

6.2.1 MVO Algorithms

Model 1: Minimize Risk Optimization

Model 1 serves to minimize risk for a given return. The objective function minimizes portfolio variance

subject to a target return and weight constraint. The structure of the model is as follows:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{x}^T Q \mathbf{x} \\ \text{s.t.} \quad & \boldsymbol{\mu}^T \mathbf{x} \geq R, \\ & \mathbf{1}^T \mathbf{x} = 1. \end{aligned}$$

Model 1 is ideal for risk-averse individuals who seek to prioritize capital preservation but want to obtain a defined return, which was calculated by taking the average of all expected returns.

Model 2: Maximize Return Optimization

Model 2 serves to maximize return while imposing an upper bound on the allowed risk. The model maximizes the portfolio return subject to a risk limit and weight constraint. The model appears as follows:

$$\begin{aligned} \max_{\mathbf{x}} \quad & \boldsymbol{\mu}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{x}^T Q \mathbf{x} \leq \epsilon^2 \\ & \mathbf{1}^T \mathbf{x} = 1 \end{aligned}$$

Contrary to Model 1, this model is ideal for investors targeting high returns but allows for safety with a predefined risk limit. This limit must indicate a balanced portfolio as very low risk will result in low expected return, but very high risk will increase volatility, both of which decrease the Sharpe ratio. A risk limit was set to 0.06, a moderate risk level, for a balanced portfolio.

Model 3: Risk-Return Trade-Off Optimization

The risk-return trade-off optimization is a combination of Model 1 and Model 2 and serves to balance both risk and expected return. The model maximizes the difference in the portfolio return and risk, where the risk level is defined by a risk-aversion parameter. The model is also subject to a weight constraint, and is structured as follows:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \mathbf{x}^T Q \mathbf{x} - \lambda \boldsymbol{\mu}^T \mathbf{x} \\ \text{s.t.} \quad & \mathbf{1}^T \mathbf{x} = 1 \end{aligned}$$

The choice of risk-aversion parameter differs by user needs. The selection of this parameter for this project is discussed more in *Section 2.2.4 Risk Aversion Parameters*.

Model 4: Robust Risk-Return Trade-Off Optimization with Ellipsoidal Uncertainty Set

Model 4 is built on Model 3 by introducing robustness to the algorithm. Robust risk-return optimization improves traditional optimization methods by addressing the issue of sensitivity to estimation errors in expected returns and the covariance matrix to make portfolios less affected by such inaccuracies. The model is as follows:

$$\begin{aligned} \min_{\mathbf{x}} \quad & \lambda \mathbf{x}^T Q \mathbf{x} - \left(\boldsymbol{\mu}^T \mathbf{x} - \epsilon \left\| \boldsymbol{\Theta}^{1/2} \mathbf{x} \right\|_2 \right) \\ \text{s.t.} \quad & \mathbf{1}^T \mathbf{x} = 1 \end{aligned}$$

A confidence level of 90% was used in the computation of epsilon, which indicates the size of the uncertainty set. A confidence level of 90% results in a larger estimation error range and allows for a balanced portfolio that manages both risk and return, which optimizes the Sharpe ratio. The testing that resulted in a confidence level of 90% is detailed in *3.5.1 Differing λ Data*.

6.2.2 Model Considerations

Model choices

Our motivation for utilizing the above models stem from the concepts learned in this course. We wanted to demonstrate our ability to successfully apply the algorithms using our practice from the lab sessions. Additionally, we wanted to make accurate and insightful conclusions to expand our knowledge of these models in a practical setting. In particular, we had an interest in robust optimization, as it is not widely used and would be a critical skill to have in the industry.

Short Selling

The goal is to maximize the portfolio Sharpe ratio while minimizing the average turnover rate, essentially crafting a theoretically optimal portfolio. The choice to allow or disallow short selling can have many implications that must be considered under the project framework. Disallowing short selling can decrease the average turnover rate but limit the Sharpe ratio due to a lack of flexibility in constructing the optimal portfolio. Conversely, allowing short selling can better maximize the Sharpe ratio due to more flexibility in weight allocations, but may increase the average turnover rate. Considering the goal of this project, and the prioritization on maximizing the Sharpe ratio, all of the models listed in *2.1 MVO Algorithms* allow short selling.

Data Cleaning and Validation

In the calculated periodic returns of the assets, there were two invalid numbers in the matrix (NaN). This was resolved by setting the invalid values in the period returns matrix to zero to avoid an error in the MATLAB optimization solver, allowing it to run successfully.

Additionally, we chose to use all five years of data rather than subsetting it to avoid excessive overfitting. Reducing the dataset to less than five years would result in a model specific to the small dataset, affecting its ability to perform on new, unseen data.

Risk Aversion Parameters

Model 3 and Model 4 in *2.1 MVO Algorithms* require a user-inputted risk aversion parameter. This parameter defines how much weight is given to minimizing risk versus maximizing return to allocate weight to such assets accordingly. A baseline risk aversion parameter of 20 was used to test the models. A parameter of 20 is within the range for a balanced portfolio, providing a good measure of the Sharpe ratio and average turnover rate. This parameter was fine-tuned once the final model was selected in *3.5.1 Differing λ Data*.

6.2.3 Regularization Techniques and Factor Models

Regularization techniques were used to determine the factors that explain the most variation in the model. These best factors were used in forming a factor model that estimates μ and Q , which were used in calculating the Sharpe ratio and average turnover rate in the MVO models. Additionally, ubiquitous factor models can be used for the same purpose. It is crucial to use factor models to estimate μ and Q because they are typically more accurate and less noisy than direct estimates. Factor models also allow for better performance on new datasets as the estimates are more generalizable. In this section, the regularization techniques used to obtain these estimates, and the reasoning behind their forms, will be introduced, in addition to well-known factor models.

Best Subset Selection (BSS)

The penalized form of BSS uses the l_0 -norm, which is structured as follows:

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \|\mathbf{B}_i\|_0$$

$$\|\mathbf{B}_i\|_0 = \mathbb{K}_{\alpha_i \neq 0} + \sum_{k=1}^p \mathbb{K}_{\beta_k \neq 0}$$

BSS selects the most important factors by setting a penalty on the number of non-zero coefficients which makes the norm values smaller. However, the penalized form does not allow explicit control on the number of non-zero elements. BSS promotes true sparsity, allowing several coefficients to be zero at optimality. However, the l_0 -norm is not convex, so BSS is not convex, resulting in BSS performing best on mixed-integer programming models with binary variables. Nevertheless, for completeness, insight, and curiosity, BSS was tested with the four algorithms listed previously.

Least Absolute Shrinkage and Selection Operator (LASSO)

The penalized form of LASSO uses the l_1 -norm and appears as follows:

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \|\mathbf{B}_i\|_1$$

$$\|\mathbf{B}_i\|_1 = |\alpha_i| + \sum_{k=1}^p |\beta_{ik}|$$

LASSO penalizes the sum of absolute values of the factors which also makes the norm values smaller. Similar to BSS, LASSO promotes true sparsity resulting in zero-valued coefficients. The l_2 -norm is a convex function, so LASSO is also convex, which allows for efficiency when solving.

Ridge Regression (RR)

RR, the final regularization technique, utilizes the l_2 -norm. The penalized form is:

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2 + \lambda \|\mathbf{B}_i\|_2^2$$

$$\|\mathbf{B}_i\|_2 = \left((\alpha_i)^2 + \sum_{k=1}^p (\beta_{ik})^2 \right)^{1/2}$$

Uniquely, RR penalizes the Euclidean norm. However, RR does not promote true sparsity and instead attempts to shrink the coefficients to zero. However, the l_2 -norm is continuous, convex, and smooth, and allows a closed-form solution.

Fama-French Three Factor Model

This Three Factor Model uses the market factor, size factor, and value factor, and is given by the following:

$$r_i - r_f = \alpha_i + \beta_{im}(f_m - r_f) + \beta_{is}f_s + \beta_{iv}f_v + \varepsilon_i$$

Note that the factors are correlated in the Three-Factor Model as they are subsets of market data.

Capital Asset Pricing Model (CAPM)

The final factor model is CAPM which is a one-factor model of the market excess return. CAPM is known by the following:

$$r_i - r_f = \beta_{im}(f_m - r_f) + \varepsilon_i$$

Ordinary Least Squares Regression (OLS)

A simple OLS regression with the eight provided factors was also used. OLS regression minimizes the sum of squared residuals, similar to BSS, LASSO, and RR, but is subject to no constraints. OLS regression is given by:

$$\min_{\mathbf{B}_i} \|\mathbf{r}_i - \mathbf{X}\mathbf{B}_i\|_2^2$$

6.2.4 Factor model considerations

The regularization techniques and factor models described above were chosen to estimate μ and Q for reasons similar to those in *2.2.1 Model Choices*. We wanted to apply a range of techniques learned in this course to develop a practical understanding of how each technique worked. This project served as an excellent opportunity for us to see the results from these methods, which expanded our understanding and built our intuition.

The regularization techniques can be used in their penalized or constrained forms. The LASSO and RR models are equivalent in their penalized or constrained form, but BSS is not. The penalized form of BSS was used because it is more flexible than its constrained form. The constrained form requires a manually selected cardinality constraint, which limits computational feasibility. Rather than arbitrarily estimating a cardinality constraint, the penalized form uses automatic factor selection which is better for large datasets where strict control is not necessary. It is better to adjust the penalty to make the norm smaller to obtain a balance between the number of factors and accuracy.

6.2.5 Assessment of Models

To properly evaluate the models, a baseline was established. This was accomplished by using the sample mean and covariance as direct estimates of μ and Q from the cleaned dataset to obtain Sharpe ratios and average turnover rates for each model, which allowed for a preliminary assessment of model performance. Furthermore, regression techniques and factor models were used to further refine the estimates of μ and Q . Sharpe ratios, average turnover rates, and runtimes were recorded for each factor model and MVO algorithm. More insightful observations were made regarding model performance. The model that achieved the largest Sharpe ratio and an acceptable average turnover rate was selected to be further optimized with parameter inputs, if applicable. The parameters were adjusted within a specified range and the optimized model was selected as best-performing based on its response to parameter changes, with close attention on the Sharpe ratio. General observations were made for each model, including a best-to-worst ranking.

6.3 Analysis

6.3.1 Baseline Data

The baseline data we began with for each model is provided in the table below.

Baseline	Model 1	Model 2	Model 3	Model 4
Sharpe Ratio	-0.001945	-0.04409	0.14652	0.16887
Average Turnover	0.073739	0.00000008	1140.35	0.85251
Runtime (s)	~2	~5	~1	~36

Table 1: Baseline data for models, with basic estimation of μ and Q from historic data.

Model 4 had the highest Sharpe ratio when using the baseline μ and Q values. While Model 3 had the second-highest Sharpe ratio, it exhibited an exceptionally high turnover rate. Therefore, Model 4 was the best initial choice.

We then began refining the optimization models by using OLS regression and factor models to estimate μ and Q . Tables 2-5 contain the baseline and refined results for all four optimization models.

6.3.2 Model 1 Data

Model 1	Baseline	OLS	CAPM	Three-Factor	BSS	LASSO	RR
Sharpe Ratio	-0.001945	0.069483	0.083962	0.07796	0.07678	0.081813	0.071452
Average Turnover	0.073739	0.2227	0.17484	0.18509	0.18383	0.18835	0.22596
Runtime (s)	~2	~1	~2	~1	~2	~2	~1

Table 2: Data collected for Model 1, Minimizing Risk Optimization.

All of the refined μ and Q estimates significantly improved the performance of Model 1. The CAPM factor model and the LASSO based factor model achieved the first and second highest Sharpe ratios, respectively. Their turnover rates were very similar, so the CAPM factor model was determined to be the best performing μ and Q estimator used in conjunction with Model 1.

6.3.3 Model 2 Data

Model 2	Baseline	OLS	CAPM	Three-Factor	BSS	LASSO	RR
Sharpe Ratio	-0.04409	0.052381	0.069067	0.063895	0.063026	0.068047	0.053804
Average Turnover	0.00000008	0.21969	0.1621	0.1738	0.17273	0.17465	0.22965
Runtime (s)	~5	~6	~7	~6	~7	~8	~6

Table 3: Data collected for Model 2, Maximizing Return Optimization.

All of the refined μ and Q estimates also significantly improved the performance of Model 2. The CAPM factor model and the LASSO-based factor model achieved the first and second highest Sharpe ratios, respectively. Their turnover rates were very similar, so the CAPM factor model was determined to be the best performing μ and Q estimator used in conjunction with Model 2.

6.3.4 Model 3 Data

Model 3	Baseline	OLS	CAPM	Three-Factor	BSS	LASSO	RR
Sharpe Ratio	0.14652	0.098093	0.099261	0.15429	0.13302	0.1222	0.1534
Average Turnover	1140.35	810.1	872.2	876.85	864.10	836.74	843.82
Runtime (s)	~1	~1	~1	~1	~1	~2	~1

Table 4: Data collected for Model 3, Risk-Return Trade-Off Optimization.

Model 3's Sharpe ratio only improved with estimates of μ and Q derived from the Fama-French three-factor model and ridge regression. The three-factor model yielded the highest Sharpe ratio, followed by ridge regression. However, all variations of Model 3 exhibited extremely high turnover rates, suggesting that its overall performance would not perform well against the assignment's evaluation criteria.

6.3.5 Model 4 Data

Model 4	Baseline	OLS	CAPM	Three-Factor	BSS	LASSO	RR
Sharpe Ratio	0.16887	0.10782	0.11163	0.11356	0.10288	0.083183	0.10463
Average Turnover	0.85251	1.45332	1.4961	1.4693	1.4577	1.6384	1.4472
Runtime (s)	~36	~38	~36	~28	~42	~69	~34

Table 5: Data collected for Model 4, Robust Risk-Return Trade-Off Optimization.

None of the factor model based estimations of μ and Q improved the performance on Model 4.

Differing λ Data

Since Model 4 achieved the highest Sharpe ratio with the baseline μ and Q estimation, we further optimized it by adjusting its risk aversion parameter, λ . We tested values from $\lambda = 15$ to $\lambda = 25$ (± 5 from $\lambda = 20$), and found λ peaked at $\lambda = 17$. Table 6 presents the results for Model 4 for $\lambda = 15$ to $\lambda = 20$.

Model 4, with lambda =	15	16	17	18	19	20
Sharpe Ratio	0.1715	0.17255	0.17272	0.1721	0.17079	0.16887
Average Turnover	1.2015	1.103	1.0228	0.95667	0.90046	0.85251
Runtime (s)	~ 38	~ 36	~ 28	~ 42	~ 69	~ 34

Table 6: Data for Model 4 with varying λ values.

The λ that achieved the highest Sharpe ratio was $\lambda = 17$. All λ 's had similar turnover rates, so $\lambda = 17$ was determined to be the optimal value of the risk aversion parameter. The confidence level, α , also demonstrated an increasing trend as it was gradually tested from 75% to 90%, where the Sharpe ratio peaked.

6.3.6 General Observations

Factor models did not always improve the performance of the optimization models (specifically Model 3 and Model 4), which was unexpected.

Since BSS typically performs best on mixed-integer programming models, it was expected to have subpar performance out of all the factor models. However, each factor model exhibited varying performance across different optimization models. As such, no specific factor model can be identified as the "best" or "worst". BSS did not perform particularly well, nor did it perform particularly poorly.

Since a model's Sharpe ratio accounted for 80% of its evaluation criteria, we prioritized maximizing the Sharpe ratio over minimizing turnover rates. Based on this approach, Model 4 was clearly the best-performing model, consistently achieving high Sharpe ratios across different factor models while maintaining acceptable turnover rates.

Although Model 3 consistently had the second-highest Sharpe ratios, its turnover rate was approximately 1,000 times higher than the other models, making it the worst-performing model. Models 1 and 2 had similar turnover rates, but Model 1 consistently achieved a higher Sharpe ratio across all factor model estimations. Therefore, Model 1 ranked as the second-best model, followed by Model 2 in third place.

6.4 Discussion

6.4.1 Strengths of Final Model

The robust risk-return trade-off optimization with ellipsoidal uncertainty had several key strengths compared to the other traditional MVO models analyzed. Unlike Models 1, 2, and 3, the ellipsoidal uncertainty set accounted for worst-case uncertainties in the asset return estimates. This reduced the risk of overfitting to historical data and made it more likely to perform well on unseen datasets. In addition, the ideal risk aversion parameter (λ), found to be 17, balanced the trade-off between minimizing risk and maximizing return effectively and optimized the portfolio's Sharpe ratio to 0.17272 for the given dataset.

Furthermore, traditional models like Model 1 and 2 are often unstable during volatile markets, so a robust optimization typically reduces turnover because it optimizes under worst-case conditions. This ensured a

more reliable performance across risky markets and is evident in the model's low turnover value of 1.0228. Lastly, our final model did meet our expectations as it was built upon the first three models. Given what we learned in the course, we anticipated that a robust optimization model would yield the best results.

6.4.2 Limitations of Final Model

The final model relied on parameters like λ and α which had to be fine-tuned to produce an optimal portfolio. Since we were tasked with automating an asset management system, finding the best parameter values requires extensive back-testing which can be difficult to automate. This is because ideal values of λ and α vary across datasets and market conditions. Although the robustness of the model has major benefits, optimizing under worst-case conditions forces the model to assume that return estimates contain uncertainty and hedges against the worst case scenario, which can lead to over-conservatism in asset management. This means that the model may miss trading opportunities when the markets are less volatile.

6.4.3 Failure of Factor Model

After incorporating various factor model selection processes to provide estimates of μ and Q for the model's objective function, they failed to produce better results than the sample mean and covariance. Possible reasons for this include that factor models assume asset returns are driven by predefined risk factors which are theoretical and not guaranteed to explain actual price movements. Many hedge funds and asset managers mine their own factors rather than relying on standard theoretical models like Fama-French. In addition, the factors given are well known and mass incorporation of them over time reduces its effectiveness and are often arbitrated away.

Furthermore, factor models estimate expected returns and covariance using historical regressions, which may fail to capture changing risk dynamics in the data, leading to poor diversification in the portfolio. On the contrary, leveraging historical mean returns and sample covariance, like we did in our trading algorithm, provided a more reliable, data-driven approach that better reflected actual market conditions. This went against our expectations, as we had learned that factor models provided stronger estimates for μ and Q than using the sample mean and covariance estimates. However, in practice, their assumptions did not hold and surprisingly, the simple approach of using historical mean returns and sample covariance performed best.

6.5 Conclusion

This project successfully developed and evaluated an automated asset management system using robust risk-return trade-off optimization with an ellipsoidal uncertainty set. The model demonstrated superior performance in maximizing the Sharpe ratio while maintaining a relatively low turnover rate. By optimizing under worst-case conditions, the model effectively balanced risk and return, outperforming traditional MVO models, particularly in volatile markets.

Despite its strengths, the model's reliance on fine-tuned parameters such as the risk aversion parameter and the confidence level introduced challenges in automation, as these parameters require extensive back-testing and may vary with market conditions. Moreover, the assumption of incorporating uncertainty into return estimates, while advantageous in reducing risk, could lead to conservative decisions that miss trading opportunities during stable market conditions.

The integration of factor models, intended to refine estimates of expected return and covariance, did not yield the expected improvements. In fact, the use of historical mean returns and sample covariance emerged as the more reliable approach. This outcome suggests that while factor models are theoretically sound, they may not fully capture the dynamic nature of asset returns, leading to suboptimal diversification and

performance in practical applications.

In future iterations, it is necessary to explore alternative approaches to factor modeling to understand why they yielded worse estimates of μ and Q than the sample mean and covariance estimates, enhancing the robustness and accuracy of the system. Nonetheless, the findings from this project highlight the importance of careful model selection, validation, and parameter tuning when developing automated asset management systems.